Aldehyde, Ketone & Carboxylic Acids

1. How can you convert Butyne to Butan2-one?

1) H_2O/H^+
2) Hg^{2+}/H_2SO_4
3) Conc. H_2SO_4/H_3PO_4
4) $K_2Cr_2O_7/KMnO_4$

2. The IUPAC name of crotonaldehyde is

1) Butanal
2) But-2-enal
3) But-1-enal
4) But-2-ene-1-al

3. $CH = CH \xrightarrow[H_2SO_4]{HgSO_4} X \xrightarrow{LiAlH_4} Y$. Are In the above reaction, X and Y are

1) Ethanal, Ethanol
2) Ethanol, Ethanal
3) Ethanal, Ethanoic acid
4) Ethanoic acid, Acetone

4. Propanoyl chloride is subjected to reduction with H_2 in boiling xylene in the presence of Pd supported by $BaSO_4$

1) Acetone
2) Propionaldehyde
3) Ethanal
4) Propanone

5. The reaction involving the preparation of salicylaldehyde from phenol using chloroform and NaOH is called

1) Gattermann reaction
2) Perkin's reaction
3) Reimer-Tiemann reaction
4) Rosenmund reaction

6. Formaldehyde when reacted with methyl magnesium bromide followed by hydrolysis gives

1) C_2H_5OH
2) CH_3COOH
3) HCHO
4) CH_3CHO

7. How would you convert Propanone to Propene?

1) $LLIAlH_2 \rightarrow$ Conc. H_2SO_4
2) DiBALH
3) Conc. $H_2SO_4 \rightarrow NaBH_4$

4) $BH_3.THF$

8. Complete the following reactions and identify A.

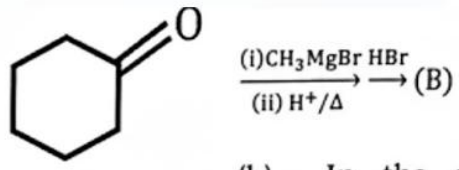
$$A + H_2(g) \xrightarrow[BaSO_4]{Pd} (CH_3)_2CH - CHO$$

1) $(CH_3)_3C-OH$

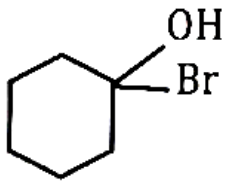
2) $(CH_3)_2CH-CHO$

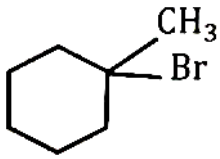
3) $(CH_3)_2CHCOCH_3$

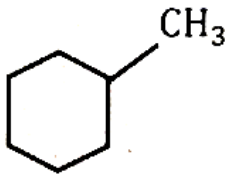
4) $(CH_3)_2CH-OH$

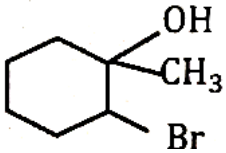


9. (b) In the above reactions is

1) 

2) 

3) 

4) 

10. Most acidic hydrogen is present in

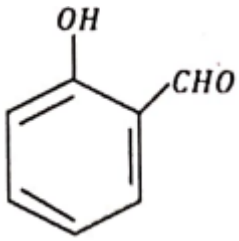
1) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$

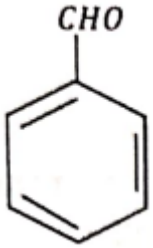
2) $CH_3-O-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_3$

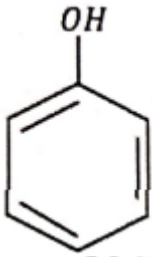
3) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_3$

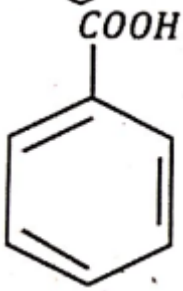
4) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_3$

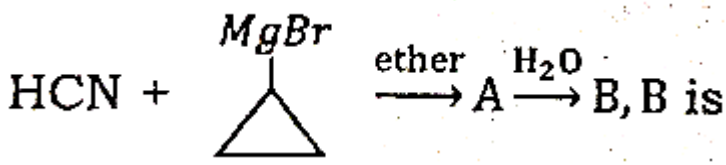
11. What will be the product, when carboxy phenol, obtained by Reimer-Tiemann's process, is deoxidized with Zn powder?

1) 

2) 

3) 

4) 

12. 

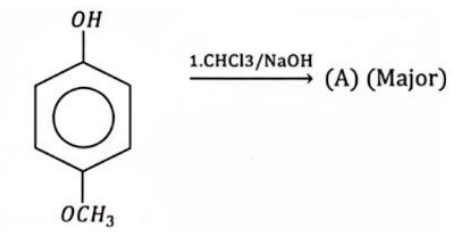
1) Cyclopropanol
2) Cyclopropanal
3) Cyclopropanecarbaldehyde
4) Cyclopropanenitrile

13. In a set of the given reactions, acetic acid yielded a product C

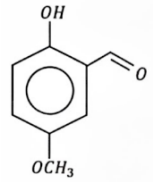
$$CH_3COOH + PCl_5 \rightarrow A \xrightarrow[\text{Anh.}AlCl_3]{C_6H_6} B \xrightarrow[\text{ether}]{C_2H_5MgBr} C$$

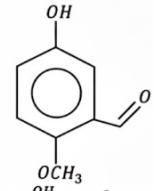
Product C would be

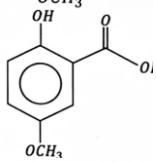
1) $CH_3CH(OH)C_2H_5$
2) $CH_3COC_6H_5$
3) $CH_3CH(OH)C_6H_5$
4) $CH_3-C(OH)(C_2H_5)C_6H_5$

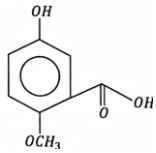


14. Product (A) is:

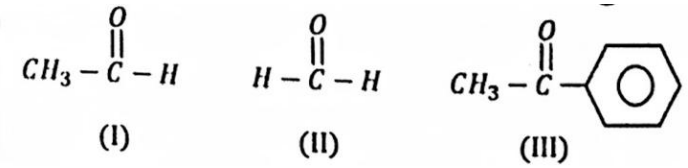
1) 

2) 

3) 

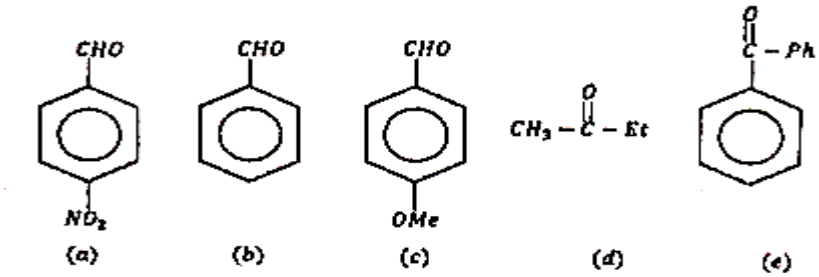
4) 

15. Correct order of reactivity of following compounds towards Grignard reagent?



1) I > II > III
2) II > I > III
3) II > III > I
4) I > III > II

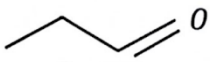
16. The correct of rate of reaction toward nucleophilic addition reaction:

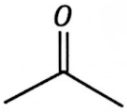


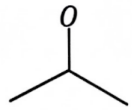
1) a > b > c > d > e
2) a > b > d > c > e
3) a > d > e > b > c
4) a > b > e > d > c

$$CH_3 - \overset{O}{\overset{||}{C}} - OH \xrightarrow[\Delta]{Ca(OH)_2} (A)$$

17. Product (A) is:

1) 

2) 

3) 

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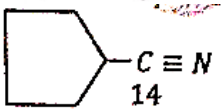
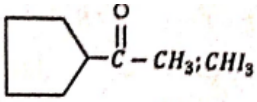
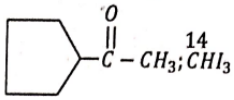
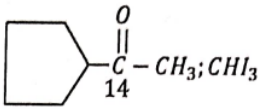


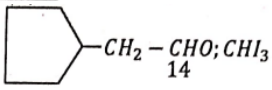
$$+CH_3MgBr \xrightarrow[H_2O^{\oplus}]{} (A) \xrightarrow{NaOI} (B) + (C)$$

18. Product (A) and (C) is:

1) 

2) 

3) 

4) 

19. Which of following does not react with $NaHSO_3$ (sodium bisulphite)?

1) $CH_3-\overset{O}{\overset{||}{C}}-H$

2) $CH_3-\overset{O}{\overset{||}{C}}-Et$

3) $Et-\overset{O}{\overset{||}{C}}-Et$

4) $Ph-CH_2-\overset{O}{\overset{||}{C}}-CH_3$

$$CH_3-CH_2-\overset{O}{\overset{||}{C}}-H \xrightarrow{HCN} (A) \xrightarrow{H_3O^{\oplus}} (B)$$
$$\xrightarrow[\Delta]{} (C) \xrightarrow{LiAlH_4} (D) \xrightarrow{HIO_4} HCHO + (E)$$

20. Compound (C) can show geometrical isomerism. Product (E) of the reaction will be:

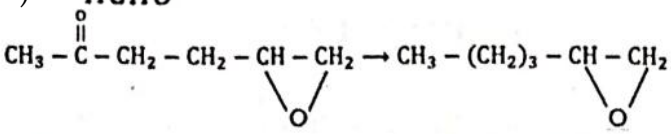
1) $CH_3-\overset{O}{\overset{||}{C}}-CH_3$

2) $CH_3-CH_2-\overset{O}{\overset{||}{C}}-H$

3) CH_3-CHO

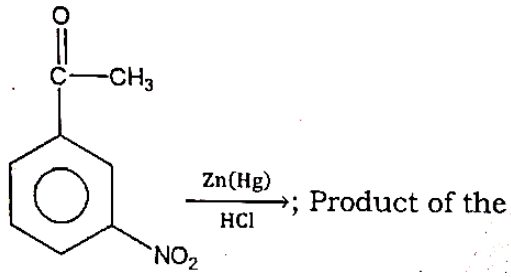
4) $HCHO$

21.

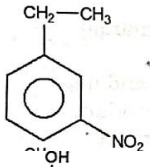


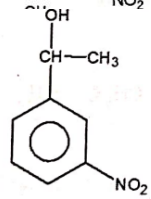
Above conversion can be achieved by:

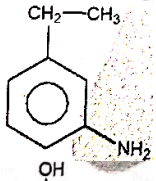
1) Wolff-Kishner reduction
2) Clemmensen reduction
3) $HS-CH_2-CH_2-SH$, following by Raney Ni
4) None of these

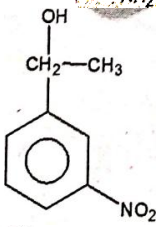
22. 

; Product of the

1) 

2) 

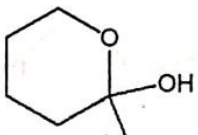
3) 

4) 

23. $CH_3-\overset{OH}{\overset{|}{CH}}-CH_2-CH_2-\overset{O}{\overset{||}{C}}-CH_3 \longrightarrow CH_3-\overset{OH}{\overset{|}{CH}}-(CH_2)_3-CH_3$

Above conversion can be achieved by:

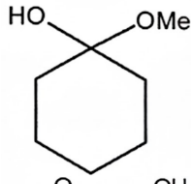
1) Wolff-Kishner reduction
2) Clemmensen reduction
3) $LiAlH_4$
4) $NaBH_4$

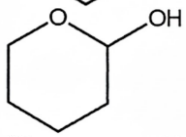
24. The structure 

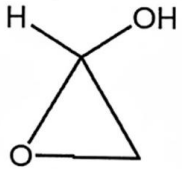
would be best classified as a/an

1) Acetal
2) Hemiacetal
3) Hydrate
4) Cyanohydrin

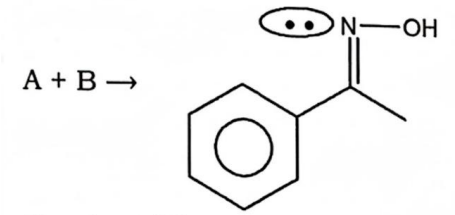
25. Which of the following compound is hemiacetal?

1) 

2) 

3) 

4) All of these



26. Reactant (A) and (B) is:

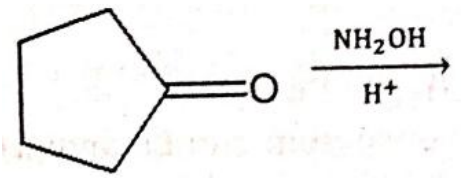
1) $Ph-CH_2-CH=O+NH_2-OH$

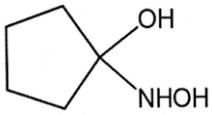
2) $Ph-CH=O+NH_2-OH$

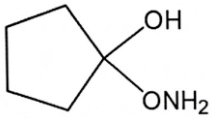
3) $Ph-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3+NH_2-NH_2$

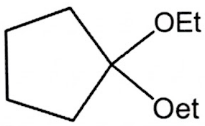
4) $Ph-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3+NH_2-OH$

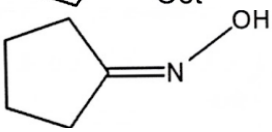
27. Which is the major product of the following reaction?



1) 

2) 

3) 

4) 

28. $Ph-\overset{OH}{\overset{|}{C}H}-CH_3 \xrightarrow{PCC} (A) \xrightarrow{NH_2-NH-\overset{O}{\overset{||}{C}}-NH_3}$ (B); Product (B) is:

1) $Ph-\overset{CH_3}{\overset{|}{C}}=N-\overset{O}{\overset{||}{C}}-NH-NH_2$

2) $Ph-\underset{CH_3}{\underset{|}{C}}=N-NH-\overset{O}{\overset{||}{C}}-NH_2$

3) $Ph-CH=N-\underset{CH_3}{\underset{|}{N}}-\overset{O}{\overset{||}{C}}-NH_2$

4) $Ph-CH=N-\overset{O}{\overset{||}{C}}-NH_2$

29. Which factors will increase the reactivity of C=O group?

(i) Presence of a group with positive inductive effect.

(ii) Presence of a group with negative inductive effect

(iii) Presence of large alkyl group

1) Only (i)
2) Both (i) and (ii)
3) Only (ii)
4) Both (ii) and (iii)

30. Nucleophilic addition reaction will be most favoured in

1) CH_3CHO

2) $CH_3-CH_2-CH_2\overset{O}{\overset{||}{C}}-CH_3$

3) $(CH_3)_2C=O$

4) CH_3CH_2CHO

Answer Key

1	2	**16**	1
2	2	**17**	2
3	1	**18**	3
4	2	**19**	3
5	3	**20**	2
6	1	**21**	2
7	1	**22**	2
8	4	**23**	1
9	2	**24**	2
10	1	**25**	4
11	2	**26**	4
12	3	**27**	4
13	4	**28**	2
14	1	**29**	3
15	2	**30**	1